

Statistical Decision Theory with Counterfactual Loss

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Goal: Evaluate the quality of decisions.

- Classical decision theory:
 - Evaluates based on *observed outcomes*.
 - Did the decision yield a successful outcome?
- This talk:
 - What happens if we use *all potential outcomes*?
 - Would a different decision have produced the same outcome? If so, would it have been preferable?

Contribution: Extend classical decision theory for treatment choice to counterfactual losses.

Wald 1950: Decision-making as a game against nature.

- 1 Nature picks an unknown state θ ,
- 2 Decision-maker chooses action $D = d$,
- 3 A **loss** $\ell(d, \theta)$ quantifies the cost of choosing d under θ .

Given covariates $\mathbf{X}$, construct a decision rule $D = \pi(\mathbf{X})$.

Measure performance with **risk**,

$$R(\pi; \theta, \ell) = \mathbb{E}_{\theta} [\ell(\pi(\mathbf{X}), \theta)].$$

Treatment Choice

Manski [2000; 2004; 2011]: Statistical decision theory for treatment choice.

Idea:

- Choose treatment $D = d$ to minimize loss based on outcome Y .
- Loss depends on potential outcome $Y(d)$, i.e., $\ell(d, Y(d))$.

Given covariates $\mathbf{X}$, use a treatment rule $D = \pi(\mathbf{X})$.

Evaluate risk

$$R(\pi; \ell) = \mathbb{E}[\ell(\pi(\mathbf{X}), Y(\pi(\mathbf{X})))]$$

Limitation: Loss only depends on the treated potential outcome.

Trichotomous Decision

Physician treating a patient

- $D = 0$: No treatment
- $D = 1$: Standard treatment (more invasive)
- $D = 2$: Experimental treatment (most invasive)

c_D cost of treatment D

Outcome

- $Y = 1$: survival
- $Y = 0$: death

ℓ_y loss under outcome $Y(D) = y$

Standard loss:

$$\ell^{\text{STD}}(D, Y(D)) = \ell_{Y(D)} + c_D$$

- $(D, Y(D)) = (1, 0) : \ell_0 + c_1$
- $(D, Y(D)) = (2, 1) : \ell_1 + c_2$

Trichotomous Decision II

Standard Loss: $\ell^{\text{STD}}(D, Y(D)) = \ell_{Y(D)} + c_D$

Clinical & ethical goal: Avoid overtreatment

- Prefer least invasive treatment that ensures survival
- Prefer option $k < d$ if $Y(k) = 1$
- r_k regret of overtreating option k

Counterfactual loss:

$$\ell^{\text{COF}}(D; Y(0), Y(1), Y(2)) = \ell_{Y(D)} + c_D + \sum_{k < D} r_k Y(k).$$

$(Y(0), Y(1), Y(2)) = (0, 1, 1)$

- $D = 1 : \ell_1 + c_1$
- $D = 2 : \ell_1 + c_2 + r_1$

We show:

- For r_k sufficiently large, ℓ^{STD} and ℓ^{COF} yield different treatment preferences.
- No standard loss that can take these ethical considerations into account.

Observed data: For each unit $i = 1, \dots, n$, observe $(\mathbf{X}_i, D_i, Y_i)$, where:

- Covariates: $\mathbf{X}_i \in \mathcal{X}$
- Decision: $D_i \in \mathcal{D} = \{0, 1, \dots, K-1\}$
- Outcome: $Y_i \in \mathcal{Y} = \{0, 1, \dots, M-1\}$
- Potential Outcome under $D = d$: $Y(d) \in \mathcal{Y}$

Aim: Study the quality of a generic decision $D_i^* \in \mathcal{D}$ (think of $D^* = \pi(\mathbf{X})$)

Assumptions:

- **Consistency:** $Y_i = Y_i(D_i)$, and if $D_i^* = D_i$, then $Y_i(D_i^*) = Y_i(D_i)$
- **Strong Ignorability:**
 - *Unconfoundedness:* $D_i \perp\!\!\!\perp (D_i^*, \{Y_i(d)\}_{d \in \mathcal{D}}) \mid \mathbf{X}_i$
 - *Overlap:* $\exists \eta > 0 : \eta < \Pr(D_i = d \mid \mathbf{X}_i) < 1 - \eta$, for all $d \in \mathcal{D}$

Counterfactual Loss and Risk

Counterfactual loss: $\ell : \mathcal{D} \times \mathcal{Y}^K \times \mathcal{X} \rightarrow \mathbb{R}$, i.e., $\ell(d; y_1, \dots, y_K, \mathbf{x})$.

Loss of choosing $D^* = d$ given

- Potential outcomes: $(Y(0), \dots, Y(K-1)) = (y_0, \dots, y_{K-1})$
- Covariates: $\mathbf{X} = \mathbf{x}$

Definition (Counterfactual Risk and Conditional Counterfactual Risk)

Given counterfactual loss ℓ , the counterfactual risk of decision D^* is:

$$R(D^*; \ell) := \mathbb{E}[\ell(D^*; Y(0), \dots, Y(K-1), \mathbf{X})] = \mathbb{E}[R_{\mathbf{X}}(D^*; \ell)]$$

where the conditional counterfactual risk given $\mathbf{X} = \mathbf{x}$ is,

$$R_{\mathbf{x}}(D^*; \ell) := \sum_{d \in \mathcal{D}} \sum_{\{y_k\}_{k=0}^{K-1} \in \mathcal{Y}^K} \ell(d; y_0, \dots, y_{K-1}, \mathbf{x}) \\ \times \Pr(D^* = d, Y(0) = y_0, \dots, Y(K-1) = y_{K-1} \mid \mathbf{X} = \mathbf{x}).$$

Problem: $\Pr(D^* = d, Y(0) = y_0, \dots, Y(K-1) = y_{K-1} \mid \mathbf{X} = \mathbf{x})$ **unidentifiable**

Identifiability of Counterfactual Risk

“Definition”: A causal parameter is identifiable if it can be expressed as a function of the observables, i.e., $\Pr(D, D^*, Y, \mathbf{X})$.

Focus on $R_{\mathbf{X}}$ (equivalent to R). Issue

$$\Pr(D^*, Y(0), \dots, Y(K-1) \mid \mathbf{X})$$

not identifiable. However, under strong ignorability

$$\Pr(D^* = d, Y(k) = y_k \mid \mathbf{X}) = \Pr(D^* = d, Y = y_k \mid D = k, \mathbf{X}).$$

Can we impose structure on ℓ that enables identification?

Definition (Additive Counterfactual Loss)

Let $\mathbf{y} = (y_0, \dots, y_{K-1}) \in \mathcal{Y}^K$. A counterfactual loss is *additive* if

$$\ell^{\text{ADD}}(d; \mathbf{y}, \mathbf{x}) = \omega_d(d, y_d, \mathbf{x}) + \sum_{k \in \mathcal{D}, k \neq d} \omega_k(d, y_k, \mathbf{x}) + \varpi(\mathbf{y}, \mathbf{x}).$$

- $\omega_d(d, y_d, \mathbf{x}) : \mathcal{D} \times \mathcal{Y} \times \mathcal{X} \rightarrow \mathbb{R}$
 - Factual weight: Contribution of observed outcome $Y(d)$
 - Decision dependent
- $\omega_k(d, y_k, \mathbf{x}) : \mathcal{D} \times \mathcal{Y} \times \mathcal{X} \rightarrow \mathbb{R}$
 - Counterfactual weight: Contribution of unobserved $Y(k)$
 - Decision dependent
- $\varpi(\mathbf{y}, \mathbf{x}) : \mathcal{Y}^K \times \mathcal{X} \rightarrow \mathbb{R}$
 - Intercept term
 - Decision independent

Examples

$$\ell^{\text{ADD}}(d; \mathbf{y}, \mathbf{x}) = \omega_d(d, y_d, \mathbf{x}) + \sum_{k \in \mathcal{D}, k \neq d} \omega_k(d, y_k, \mathbf{x}) + \varpi(\mathbf{y}, \mathbf{x}).$$

Recall the example: $\mathcal{D} = \{0, 1, 2\}$ and $\mathcal{Y} = \{0, 1\}$

Standard loss:

$$\ell^{\text{STD}}(D, Y(D)) = \ell_{Y(D)} + c_D$$

Counterfactual loss:

$$\ell^{\text{COF}}(D; Y(0), Y(1), Y(2)) = \ell_{Y(D)} + c_D + \sum_{k < D} r_k Y(k)$$

Weights	Factual $\omega_d(d, y_d, \mathbf{x})$	Counterfactual $\omega_k(d, y_k, \mathbf{x})$	Intercept $\varpi(\mathbf{y}, \mathbf{x})$
ℓ^{STD}	$\ell_{y_d} + c_d$	0	0
ℓ^{COF}	$\ell_{y_d} + c_d$	$r_k y_k \mathbf{1}\{k < d\}$	0

Both are additive

Non-Additive Loss

Same setting but only two treatments: $\mathcal{D} = \{0, 1\}, \mathcal{Y} = \{0, 1\}$.

Principal strata:

- Never survivors: $(Y(0), Y(1)) = (0, 0)$
- Responders: $(Y(0), Y(1)) = (0, 1)$
- Harmed: $(Y(0), Y(1)) = (1, 0)$
- Always survivors: $(Y(0), Y(1)) = (1, 1)$

Assign different losses to each principal strata (Ben-Michael, Imai, and Jiang 2024):

$$\begin{aligned}\ell(D; Y(0), Y(1)) = & (1 - Y(0))Y(1)\ell_D^R + Y(0)(1 - Y(1))\ell_{1-D}^H \\ & + Y(0)Y(1)\ell_1 + (1 - Y(0))(1 - Y(1))\ell_0 + c_D,\end{aligned}$$

Hippocratic Oath — “Do no harm”: Causing harm with treatment is worse than failing to provide treatment. Asymmetry in loss:

$$\underbrace{\Delta^R = \ell_0^R - \ell_1^R}_{\text{Failure to treat a responder}} < \underbrace{\Delta^H = \ell_0^H - \ell_1^H}_{\text{Harming a patient}}.$$

Non-additive loss if $\Delta^R \neq \Delta^H$.

Additivity Implies Identifiability

Theorem (Additivity Implies Identifiability)

Let ℓ^{ADD} be additive. Then,

$$R(D^*; \ell^{\text{ADD}}) = \sum_{d \in \mathcal{D}} \sum_{k \in \mathcal{D}} \sum_{y \in \mathcal{Y}} \mathbb{E}[\omega_k(d, y, \mathbf{x}) \Pr(D^* = d, Y(k) = y \mid \mathbf{X})] + \mathbb{E}[C(\mathbf{X})],$$

where

$$C(\mathbf{x}) = \sum_{\mathbf{y} \in \mathcal{Y}^K} \varpi(\mathbf{y}, \mathbf{x}) \Pr(\mathbf{Y}(\mathcal{D}) = \mathbf{y} \mid \mathbf{X} = \mathbf{x}),$$

with $\mathbf{Y}(\mathcal{D}) = (Y(0), \dots, Y(K-1))$.

Decomposition into identifiable marginal term and unidentifiable term, not depending on D^* .

Thus an additive loss yields an identifiable risk (up to a constant).

Additivity is Necessary and Sufficient

Can a counterfactual risk be identified under a non-additive loss?

Theorem

Under strong ignorability, the counterfactual risk $R(D^; \ell)$ is identifiable (up to a constant) if and only if the loss ℓ is additive.*

Binary Case: Accuracy

Consider $Y \in \{0, 1\}$, omit covariates and intercept.

$$\ell^{\text{ADD}}(d; \mathbf{y}) = \omega_d(d, y_d) + \sum_{k \neq d} \omega_k(d, y_k)$$

Let $Y = 1$ be desired, $Y = 0$ undesired. Consider:

$$\omega_d(d, 1) \leq \{\omega_k(d, 0)\}_{k \neq d} \leq 0 \leq \{\omega_k(d, 1)\}_{k \neq d} \leq \omega_d(d, 0)$$

Outcome	Decision: $D^* = d$
$Y(d) = 0$	False negative $\omega_d(d, 0)$
$Y(d) = 1$	True positive $\omega_d(d, 1)$

$\omega_d(d, y)$ accounts for **accuracy**.

Binary Case: Difficulty

$$\ell^{\text{ADD}}(d; \mathbf{y}) = \omega_d(d, y_d) + \boxed{\sum_{k \neq d} \omega_k(d, y_k)}$$

$$\omega_d(d, 1) \leq \{\omega_k(d, 0)\}_{k \neq d} \leq 0 \leq \{\omega_k(d, 1)\}_{k \neq d} \leq \omega_d(d, 0)$$

Outcome	Decision: $D^* = d$
$Y(k) = 0$	Avoid negative $\omega_k(d, 0)$
$Y(k) = 1$	Miss positive $\omega_k(d, 1)$

Rewards *consequential* decisions that change outcomes.

Let $D^* = 0$. Consider units

- $(Y(0), Y(1), Y(2)) = (1, 0, 0)$: $\omega_0(1, 1) + \omega_1(1, 0) + \omega_2(1, 0)$
 - Difficult Decision: Large loss reduction
- $(Y(0), Y(1), Y(2)) = (1, 1, 0)$: $\omega_0(1, 1) + \omega_1(1, 1) + \omega_2(1, 0)$
 - Easier Decision: Small loss reduction

$\omega_k(d, y)$ accounts for **difficulty**

Standard decision theory (Manski) can account for accuracy, but not difficulty!

Binary Outcomes: Result

Corollary

Assume $Y \in \{0, 1\}$. Let ℓ^{ADD} be additive. Then,

$$\begin{aligned} R_{\mathbf{x}}(D^*; \ell^{\text{ADD}}) &= \sum_{d \in \mathcal{D}} \zeta_d(d, \mathbf{x}) \Pr(D^* = d, Y(d) = 1 \mid \mathbf{X} = \mathbf{x}) \\ &\quad + \sum_{d \in \mathcal{D}} \sum_{k \in \mathcal{D}, k \neq d} \zeta_k(d, \mathbf{x}) \Pr(D^* = d, Y(k) = 1 \mid \mathbf{X} = \mathbf{x}) \\ &\quad + \sum_{d \in \mathcal{D}} \xi(d, \mathbf{x}) \Pr(D^* = d \mid \mathbf{X} = \mathbf{x}) \\ &\quad + C(\mathbf{x}). \end{aligned}$$

where $\zeta_k(d, \mathbf{x}) = \omega_k(d, 1, \mathbf{x}) - \omega_k(d, 0, \mathbf{x})$, $\xi(d, \mathbf{x}) = \sum_{k \in \mathcal{D}} \omega_k(d, 0, \mathbf{x})$, and

$$C(\mathbf{x}) = \sum_{\mathbf{y} \in \{0,1\}^K} \varpi(\mathbf{y}, \mathbf{x}) \Pr(\mathbf{Y}(\mathcal{D}) = \mathbf{y} \mid \mathbf{X} = \mathbf{x}).$$

Decomposition into **accuracy**, **difficulty**, **decision** and **unidentifiable** constant term.
Choosing weights

$$\omega_d(d, 1) \leq \{\omega_k(d, 0)\}_{k \neq d} \leq 0 \leq \{\omega_k(d, 1)\}_{k \neq d} \leq \omega_d(d, 0),$$

yields $\zeta_d(d, \mathbf{x}) \leq 0 \leq \zeta_k(d, \mathbf{x})$ for all $k \neq d$. Implies risk decreases with accuracy and increases with counterfactual regret.

Additive Loss and Standard Loss I

Question: Is ℓ^{ADD} a generalization of ℓ^{STD} ?

- Can standard losses replicate decision-behavior from additive losses?
- i.e. for ℓ^{ADD} exist ℓ^{STD} with $R(D^*; \ell^{\text{ADD}}) - R(D^*; \ell^{\text{STD}})$ constant in D^* ?

Proposition

If $\mathcal{D} = \{0, 1\}$, any additive counterfactual loss $\ell^{\text{ADD}}(d; y_0, y_1, \mathbf{x})$ is equivalent to a standard loss $\ell^{\text{STD}}(d, y_d)$ such that the risk difference

$$R(D^*; \ell^{\text{ADD}}) - R(D^*; \ell^{\text{STD}})$$

does not depend on D^ .*

If decisions are binary, should we even use additive counterfactual losses?

- ℓ^{STD} : no strata-based ($Y(0), Y(1)$) interpretation.
- ℓ^{ADD} : interpretable via principal strata.

Proposition

Suppose $|\mathcal{D}| \geq 3$. Then, for any additive counterfactual loss with at least one real counterfactual weight $\omega_k(d, y_k, \mathbf{x})$ there exists **no** standard loss $\ell^{\text{STD}}(d; y_d)$ such that the risk difference

$$R(D^*; \ell^{\text{ADD}}) - R(D^*; \ell^{\text{STD}})$$

does not depend on D^* .

Next steps and extensions:

- Incorporating time-dependent decisions and outcomes
- Relaxing strong ignorability
- Continuous outcomes Y

Thank you!

Happy to talk counterfactuals: What should I have done?

Scan the QR code to view the paper.



References I

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